

ST211: Applied Regression 2024/25

End-of-Term Project: Group Project

What Life Factors are Associated with Higher Incomes at 25? Do Disabilities, Learning Difficulties, Ethnicity, and Sex Affect This?

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Introduction

This report has been commissioned by the Department for Work and Pensions (DWP) and the Department for Education (DfE) for our statistical consultancy, GovernData Analytics, to understand the life factors associated with higher incomes at 25, and whether disabilities, learning difficulties, ethnicity, or sex specifically impact this. The dataset is a subset of longitudinal survey data from the DWP and DfE, containing lifestyle, socio-economic, health, and demographic information for 5792 people in England for eight waves of their lives, from 2004 to 2015/16. The Office of National Statistics (ONS) has illustrated how information, particularly regarding childhood or young-life circumstances significantly affects future income (ONS, 2016). Furnham and Cheng (2013) echoed these findings, using the National Child Development Study to show that adult earnings are significantly predicted by childhood intelligence, educational qualifications, and current occupational status. As the data supports the findings of these studies, this dataset is well-suited to answer the question at hand. Therefore, a regression analysis was conducted on a cleaned sample of 3,344 individuals to identify the significant factors that lead to higher continuous weekly incomes at 25 as:

1. Wave 1: Main Parent's Current Employment Status
2. Wave 1: Whether the Main Parent has a Long-Term Illness, Disability, or Infirmary
3. Wave 1: Mother's Highest Qualification
4. Wave 1: Mother's Marital Status
5. Wave 1: Family's National Statistics Socio-Economic Classification (NS-SEC)
6. Wave 1: Young Person's Self-Identified Ethnic Group (Wave 1)
7. Wave 1: Likelihood of the Young Person (YP) Applying to University
8. Wave 1: Whether the Young Person has a Long-Term Disability or Health Condition
9. Wave 4: Whether the Young Person has Ever Tried Cannabis
10. Wave 5: Whether the Young Person is Currently in an Apprenticeship
11. Wave 6: Whether the Young Person is Currently Attending University
12. Wave 6: Activity Status of Young Person Not in Education, Employment, or Training
13. Wave 8: Young Person's Gender

Exploratory Analysis

We used box plots and scatter plots to gain preliminary insights into potential predictors for our continuous outcome variable, *weeklyinc*. This approach helped diminish previously held intuitions about potential predictor variables – academia was a category of predictors that was assumed to significantly affect continuous weekly income, and surprisingly variables like *indschool* (whether the individual attended private) and *uniapp* (likelihood of applying to university) did not seem to have much effect, however *hiqualW6* (highest academic qualification) did, in line with our intuition. As expected, socio-economic factors such as *NSSEC* (NS-SEC class), *tenureHH* (household tenure), and variables surrounding household structure (e.g. *singleparent*, showing single parent households, and *youngsib* as the number of younger siblings) affected *weeklyinc*. Additionally, while marital status (*maritalCM*) did not affect the outcome, having a child did (*haschildW4*, *haschildW6*). *backact* (backcoded employment activity) also seemed to have a large effect. Overall, these findings aligned with our intuition as the outcome variable is a socio-economic factor, however, *finmanage* (financial management) did not seem to have an effect, but this could be due to its encompassing money management and habits, not necessarily personal income level.

Furthermore, it was revealed that health and lifestyle factors did not much affect continuous weekly income at 25, including substance use (e.g. *canntry*, showing if the YP has tried cannabis and *alcfreq*, alcohol frequency) and psychological or physical illnesses (e.g. *depress*, displaying depression, and *illnessYP*, if the YP has any long standing illnesses or disabilities). We did find that demographic factors, such as ethnicity (*ethnicity*) and treatment due to ethnicity (*teachdisc*), showed quite a large effect, with white participants earning the highest. It also showed gender (*sex*) as having no effect.

Analysis

Approach I: Backwards Elimination

We built a model through backwards elimination to ensure each predictor was investigated. With all predictors, we predicted multicollinearity between *jobstatusM* and *fulltimeM*, and similarly *jobstatusD* and *fulltimeD*, as these convey redundant information regarding parental work status. This was addressed by running VIF analysis on these variables, and other categories to avoid aliased coefficients. Any predictors which exhibited multicollinearity were removed from the in favour of predictors with less levels, alongside other non-significant predictors at the 5% level. Using ANOVA, we assessed whether categorical predictors were wholly significant, and any non-significant predictors were removed. Utilising diagnostic plots and a histogram of standardised residuals, we observed balanced Q-Q residuals, with slight skew at the lower end. The histogram supported this due to its roughly symmetrical, bell shape with slight positive skew. The funnel shape of the residuals vs. fitted plot indicated heteroskedasticity, so a log transformation was performed on the outcome variable, *weeklyinc*, improving the R^2 . Again, more non-significant variables were removed.

We investigated three possible interactions: *NSSEC* and *workstatusMP* to explore if the effect of class on income varies depending on whether the main parent is employed, *illnessMP* and *workstatusMP* to see whether employment status influences the impact of the main parent's health on income, and *sex* and *canntry* to see if the relationship between cannabis use and later income differs by gender, reflecting gendered patterns. Each interaction increased the R^2 value of the model (i.e. improved how well the model fits the data), but ANOVA proved these interactions to be non-significant, and so no interactions were used moving forwards.. Consequently, we moved on to identify outliers, as their presence was indicated from the diagnostic plots, using a function of standardised residuals, leverage, Difference in Fits (DFFITS), and Cook's distance. Where data points which simultaneously exhibited high leverage and high influence (which could both distort regression estimates and inflate standard errors if not dealt with) were removed from the dataset.

In a final attempt to increase the performance of the model, we replaced the artificial "Missing" category with NA, as had been created in data cleaning, to assess whether the inclusion of these data points impacts the regression significantly. This resulted in a decrease in the R^2 value from 0.798 to 0.791. Given the minimal change, we decided to keep the "Missing" category in the final model, as it does not significantly alter the model's fit, and allows for all variables in the final model to be kept. Lastly, we validated the performance of this model through cross-validation by fitting the model on three training/test sets: 50%, 70%, and 90%). The model's performance was relatively consistent between the splits however, the model did have a tendency to predict higher incomes. Unfortunately, further techniques such as MSPE were unavailable to be completed (see Appendix B).

Approach II: Forward Selection

We built a second model using forward selection, where the first iteration used predictors we thought may be significant, from the exploratory data analysis. This showed 13 non-significant predictors at the 5% level, which were removed. At this point, VIF analysis showed no multicollinearity. We used diagnostic plots and a histogram of standardised residuals which presented similar results to the backwards selection model, including heteroskedasticity in the residual vs. fitted plot. Therefore, we implemented a log transformation of the outcome variable which improved our R^2 value, and we then investigated three possible interactions between *tenureCM* and *backact* to examine whether the participant's economic activity influences the association between housing stability and income, *schoolW5* and *schoolW6* to explore whether continuing in education or not interact to influence later income, and *workstatusMP* and *NSSEC*, as in the backwards selection model, to explore whether main parent's employment status influences the effect of class on income. All of these interactions increased the R^2 , but were not significant at the 5% level (using ANOVA), and the interaction of *tenureCM* and *backact* introduced aliased coefficients, so no interactions were included in the model.

We used standardised residuals, leverage, DFFITS, and Cook's distance to identify outliers that exceeded typical thresholds. We found no observations had a Cook's distance exceeding 1, however several observations exceeded the DFFITS and leverage thresholds, meaning they had an unusual combination of predictor variables. As these can disproportionately influence the model, we dropped observations that exceeded both thresholds. This removed (41) points that were structurally unusual and had a large influence over the model from the dataset. Again, we replaced the artificial "Missing" category with NA to assess how this category impacts the regression. This resulted in a decrease in the R^2 value from 0.7544 to 0.71, and so this category remained. Unfortunately, cross-validation could not be performed for this model, so its predictive accuracy and MSPE remain unclear (Appendix B).

The Final Model

Our first approach (using backwards selection) resulted in better summary statistics, detailed below.

Results

Model Summary Statistics

Model Statistic	Value
Residual Standard Error	0.108
Multiple R-squared	0.798
Adjusted R-squared	0.7956
F-statistic	340 on 38 and 3271 DF
p-value	< 2.2e-16

Regression Coefficients and Significance Levels

Variable	Estimate	Std. Error	Pr(>t)
(Intercept)	5.215446	0.023072	< 2e-16
workstatusMPEmployed	0.156056	0.005332	< 2e-16
workstatusMPFull Time Education	-0.036140	0.025390	0.1547
workstatusMPOther	-0.056527	0.041413	0.1724
workstatusMPSick/Disabled	0.032213	0.013853	0.0201
workstatusMPUnemployed/Seeking Work	0.001841	0.017426	0.9159
illnessMP2	0.020796	0.005026	3.60e-05
hiquaIMApprentice/Vocational Training	0.095381	0.009370	< 2e-16
hiquaIMFurther Education	0.089996	0.007235	< 2e-16
hiquaIMGCSE	0.111712	0.006143	< 2e-16
hiquaIMHigher Education	0.082920	0.006752	< 2e-16
maritalM2	0.160707	0.009897	< 2e-16
maritalM3	0.070672	0.012953	5.23e-08
maritalM4	0.024571	0.013707	0.0731
maritalM5	0.011864	0.011631	0.3078
maritalM6	0.004697	0.018158	0.7959
maritalM7	-0.016864	0.077461	0.8277
NSSECIntermediate/Skilled	0.112517	0.012281	< 2e-16
NSSECProfessional/Managerial	0.129787	0.012513	< 2e-16
NSSECRoutine/Manual	-0.014050	0.012120	0.2465
ethnicity2	-0.256872	0.009830	< 2e-16
ethnicity3	-0.255226	0.008377	< 2e-16
ethnicity4	-0.251691	0.010120	< 2e-16

ethnicity5	-0.247866	0.012381	< 2e-16
ethnicity6	-0.264589	0.013782	< 2e-16
ethnicity7	-0.258169	0.014325	< 2e-16
ethnicity8	-0.251528	0.014347	< 2e-16
uniapp2	-0.003670	0.004343	0.3982
uniapp3	-0.032558	0.006279	2.28e-07
uniapp4	-0.067338	0.008582	5.77e-15
illnessYP2	0.081558	0.011012	1.64e-13
illnessYP3	0.091853	0.009125	< 2e-16
canntryYes	0.045337	0.004270	< 2e-16
apprentW52	0.054643	0.009260	3.99e-09
uniW62	-0.042834	0.004238	< 2e-16
NEETact2	-0.058386	0.025977	0.0247
NEETact3	-0.052867	0.026310	0.0446
NEETactMissing	0.079250	0.010629	1.14e-13
sexMale	-0.085625	0.003854	< 2e-16

Comments About the Data and Analysis

On Missingness

In maintaining the “Missing” category, we thought it necessary to investigate missingness further. The data categorises missing responses as either interviewee non-response, such as the cohort member refusing to answer, or from survey/interviewer issues, such as incomplete surveys. We predicted the majority of missing data to be interviewee non-response, or attributable to biases like social desirability bias. We speculated that these biases could have brought about some counterintuitive associations, for instance, *canntry* (whether the respondent had ever tried cannabis) showed a positive association with income, as individuals who had tried cannabis reported higher incomes. We speculate that this is due to the underreporting of the use of cannabis by individuals with lower incomes due to social desirability bias. Investigating the original dataset showed that for father-related variables, the cause is the father not being present, but broadly, missingness appears to be due to inapplicable questions. Implementing a binary measure of data presence allowed us to determine that missingness is predictive of the outcome. In *hiqualD*, *fulltimeD*, *jobstatusD*, *annualincMP*, *annualincHH*, *debt*, and *alcfreq*, missingness is broadly associated with lower weekly income, while missingness in

haschildW4, *schoolW6*, and *NEETact* corresponds with higher income. If missing data is associated with underrepresented or stigmatised backgrounds, excluding these cases biases the regression results.

We examined these characteristics in a bid to uncover patterns in missing entries. We found father-related variables are disproportionately missing for participants from single parent households and lower NS-SEC classes, as well as those from Black African, Black Caribbean, and mixed ethnic backgrounds, while variables regarding alcohol, salary, and debt are often missing among Pakistani and Bangladeshi respondents. This pattern is also seen in lower NS-SEC classes. As these patterns show missing entries to be common among certain groups, observed effects of these variables in the model could be dampened, and so should be interpreted with caution and understanding of this fact.

Do Disabilities, Learning Difficulties, Ethnicity, Class, or Sex Affect This?

We understand disabilities, learning difficulties, ethnicity, socioeconomic class and sex to affect whether a participant saw higher earnings at 25 as variables reflecting these classifications were significant in our model. Firstly, disabilities, of the participant and of their main parent, were seen to have a negative impact on higher income in Wave 8. Those whose parent did not have a disability at Wave 1 earned, on average, about 2.1% more than those whose parent did have a disability, holding everything else constant. Additionally, participants with no disability, or one which did not affect school saw 9.2% and 8.2% increases in income respectively, highlighting this as an important factor. We can also conclude from these results that learning difficulties affect income at 25, as those with disabilities which impacted school are in the baseline category, experiencing lower earnings.

Our model shows socioeconomic status to be indicative of income, as it contains many variables pertaining to this (e.g. *NSSEC*, *workstatusMP*). Overall, the 'Routine/Manual' NS-SEC class is the only level of this variable which details an income decrease of 1.5%, and as all other levels see increases in future income, with Unemployed/Inactive as the baseline. Additionally, it is shown that participants from ethnic minority backgrounds earn at least 25% less than their white surveyed counterparts. Our model shows those of Black Caribbean descent to be negatively impacted more so than any other ethnic group, with a 26% decrease in income. These results align with real-world disparities in income. In contrast, we see sex affect income with a result that diverges from established understandings of the gender pay gap as the model reports women to earn 8.6% more than men.

The Lay Report

The following analysis was done to aid the Department for Work and Pensions (DWP) and the Department for Education (DfE) in order to understand which circumstances and characteristics in early life affect income levels by the time individuals turn 25. Our findings were ultimately based on a reduced sample of 3,300 English participants who took part in a long-term study from adolescence to early adulthood. This study prominently tracked family background, education, health, and other factors that may affect the trajectory of their lives. The central question of our analysis was:

What Life Factors are Associated with Higher Incomes at 25?

The Findings

Our analysis explains around 80% of the income difference at 25. Our key findings are as follows.

1. **Parental Situation is Important**
We observed that when young people had a main parent that was not working, either due to retirement, illness, or unemployment, they had a lower income at 25 compared to other young adults whose main-parent was employed full-time, heavily affecting the results.
2. **Experience Matters**
Individuals in an apprenticeship or attending university around the ages of 20-21 had higher income levels at 25. We can interpret this as these individuals' academic and workplace experience making them desirable candidates in the workplace, securing higher income jobs.
3. **Socio-Economic Classification, Ethnicity and Disability are Strong Indicators of Income**
This model has shown higher incomes for those from high NS-SEC classes, of white ethnic background, who do not identify as disabled. The effects of these on income vary from 8% to 26%, and the cumulative effect for those who have intersectional identities would be larger.
4. **The Impact of Gender on Income Favours Women (in this case)**
We see women earn 8.6% more than men. This result suggests gender disparity in income that favours women in this cohort, diverging from societal understandings of the gender pay gap.

The Implications of Our Findings

These findings offer several insights for government policy. First, our observations show that support for disadvantaged families is vital in ensuring young people's economic futures. Therefore, we recommend heightened support for unemployed parents to tackle the impact long-term unemployment has on the next generation's level of income. We also recommend young adults should be encouraged to apply for a university degree, or gain practical experience in the workforce. This could be encouraged through the provision of scholarships, outreach programmes, and highlighting alternative venues for further education, as UCAS (2023) highlights apprenticeships are in high demand. All the while, we acknowledge that disparities in income between ethnic groups and gender requires a broader policy response beyond education or employment access. Furthermore, our findings which convey an inverse gender pay gap require further investigation as our analysis on a sample of participants should not be generalised. Finally, while the model strongly determines variables which demonstrate influence in future income, effects such as parental education and reported household income may be larger than we could estimate as families with lower income levels did not report such information. Thus, we finally recommend future studies make greater efforts to sample participants from underrepresented backgrounds to ensure equal data representation, and highlight the importance of robust sampling techniques to reduce response, self-reporting, or other biases to ensuring accurate and reliable reporting of socioeconomic factors and outcomes.

Bibliography

- Furnham, A. and Cheng, H., 2013. Factors influencing adult earnings: Findings from a nationally representative sample. *The Journal of Socio-Economics*, 44, pp.120–125. Available at: <https://doi.org/10.1016/j.socec.2013.02.008>.
- Goldberg, D.P. et al., 1997. The validity of two versions of the GHQ in the WHO study of mental illness in general health care. *Psychological Medicine*, 27(1), pp.191–197. Available at: <https://doi.org/10.1017/S0033291796004242>.
- Office for National Statistics, 2016. How do childhood circumstances affect your chances of poverty as an adult? [online] Available at: <https://www.ons.gov.uk/peoplepopulationandcommunity/educationandchildcare/articles/howdochildhoodcircumstancesaffectyourchancesofpovertyasanadult/2016-05-16>
- UCAS and Sutton Trust, 2023. *Three in five do not pursue apprenticeships because they cannot find one, say UCAS and Sutton Trust*. [online] 6 July. Available at: <https://www.ucas.com/corporate/news-and-key-documents/news/three-five-do-not-pursue-apprenticeships-because-they-cannot-find-one-say-ucas-and-sutton-trust#:~:text=This%20comes%20amid%20a%20significant,from%20the%20most%20disadvantaged%20backgrounds>

Appendix

Appendix A: Merging and Releveling

We merged the levels of 19 variables. While a majority of our merging was based on natural ordering, we looked to the literature to merge variables reporting GHQ-12 scores into a binary variable, as justified by Goldberg et al. (1997), due to their function as a case detector and for practicality reasons. Additionally, as the data cleaning resulted in “Missing” categories, these are kept as separate levels for the categorical variables where these were implemented. All merged levels are detailed below, with the baseline level starred (*).

workstatusMP

1, 2, 3, 4 → Employed
5, 7 → Unemployed/Seeking Work
6 → Full Time Education
8, 9 → Sick/Disabled
10, 11 → Retired/Not Seeking Work
12, Other, NA → Other*

hiqualM

1, 2, 3, 4, 5, 6 → Higher Education
7, 8, 9, 10, 11, 12 → Further Education
13, 14, 17, 18 → Apprentice/Vocational Training
15, 16 → GCSE
19, 20, NA → Other/Unspecified*

hiqualD

1, 2, 3, 4, 5, 6 → Higher Education
7, 8, 9, 10, 11, 12 → Further Education
13, 14, 17, 18 → Apprentice/Vocational Training
15, 16 → GCSE
19, 20 → Other/Unspecified *
Missing → Missing
NA → NA

jobstatusM

1, 2 → Employed
3 → Unemployed/Seeking Work
4, 5 → In Education/Training
6, 7 → Retired/Not Seeking Work *
8 → Sick/Disabled
9, NA → Other

jobstatusD

1, 2 → Employed
3 → Unemployed/Seeking Work
4, 5 → In Education/Training
6, 7 → Retired/Not Seeking Work
8 → Sick/Disabled
9 → Other *
Missing → Missing
NA → NA

youngsib

0 → None *
1 → One
2 → Two
3 → Three
4, 5, 6, 7, 8, 9, NA → Four+

depkids

1 → One*
2 → Two
3 → Three
4, 5, 6, 7, 8, 9, 10, NA → Four+

infants

0 → None*
1 → One
2, NA → Two

preteen

0 → None*
1 → One
2 → Two
3, 4, 5, NA → Three+

tween

0 → None*
1 → One
2 → Two
3, 4, NA → Three+

teen

0 → None*
1 → One
2, NA → Two

NSSEC

1, 2 → Professional/Managerial
3, 4, 5 → Intermediate/Skilled
6, 7 → Routine/Manual
8, NA → Unemployed/Inactive*

ghq12W2 (Goldberg et al., 1997)

≥ 3 → 1 (Poor mental health)
< 3 → 0 (Good mental health)*
NA → NA

hiqua1W6

1, 2 → Higher Education
3, 4 → AS/A-level
5, 6 → GCSE
7, 8, 9, NA → Other/None*

alcfreq

1, 2 → Frequent
3, 4 → Occasional
5, 6 → Rare*
Missing → Missing
NA → NA

jobstatusYP

1, 2 → Employed
3 → Unemployed/Seeking Work
4, 5 → In Education/Training
6, 7 → Retired/Not Seeking Work
8 → Sick/Disabled
9, NA → Other*

ghq12W8 (Goldberg et al., 1997)

≥ 3 → 1 (Poor mental health)
< 3 → 0 (Good mental health)*
NA → NA

currentact

1, 2, 3, 4 → Employed
5 → Unemployed
6, 7, 8, 12 → In Education/Training
9, 10 → Sick/Disabled
11 → Unpaid/Volunteer Work
13 → Not Seeking Work*
14, NA → Other

backact

1, 2 → Employed
3 → Unpaid/Volunteer Work
4 → Unemployed
5, 6, 7 → In Education/Training
8 → Sick/Disabled
9 → Not Seeking Work*
10, NA → Other

Appendix B: Cross-Validation

We successfully plotted the Predicted vs. Original and Predicted vs. Error plots for our backwards elimination model, however this posed difficulty with generating cross-validation plots for the forwards selection model and calculating the MSPE for each model, and comparatively. We have detailed the approaches attempted below, with all code in our .txt file.

1. Considering Merging

We came to the conclusion that we could not justify merging the levels of these values as variables had neither natural-ordering or box plot similarity.

2. Creating a Dataset with Rare Values

This approach was attempted to ensure such a value is always included in the training set, but was unsuccessful in implementation, following how we were unable to justify merging, there were multiple rare values which caused issues.

3. Using set.seed()

Using set.seed(number) to vary the number and then generate training and test sets for different seeds until there is a combination that is successful, did not work. As this is a long, iterative process, we did not reach success.